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Genetic and Pharmacological Inhibition of Retinoic Acid Receptor γ Function Promotes Endochondral Bone Formation

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Abstract

The nuclear retinoic acid receptors (RARs) play key roles in skeletal development and endochondral ossification. Previously we showed that RAR γ regulates chondrogenesis and that pharmacological activation of RAR γ blocked heterotopic ossification (HO), pathology in which endochondral bone forms in soft tissues. Thus, we reasoned that pharmacological inhibition of RAR γ should enhance endochondral ossification, leading to a potential therapeutic strategy for bone deficiencies. We created surgical bone defects in wild type and RAR γ -null mice and monitored bone healing. Fibrous, cartilaginous and osseous tissues formed in both groups by day 7, but more cartilaginous tissue formed in mutants within and around the defects compared to controls. Next, we implanted a mixture of Matrigel and rhBMP2 subdermally to induce ectopic endochondral ossification. Administration of RAR γ antagonists significantly stimulated ectopic bone formation in wild type but not in RAR γ -null mice. The antagonist-induced increases in bone formation were preceded by increases in cartilage formation and were accompanied by higher levels of phosphorylated Smad1/5/8 (pSmad1/5/8) compared to vehicle-treated control. Higher pSmad1/5/8 levels were also observed in cartilaginous tissues forming in healing bone defects in RAR γ -null mice, and increases in pSmad1/5/8 levels and *Id1-luc* activity were observed in RAR γ antagonist-treated chondrogenic cells in culture. Our data show that genetic or pharmacological interference with RAR γ stimulates endochondral bone formation and does so at least in part by stimulating canonical BMP signaling. This pharmacologic strategy could represent a new tool to enhance endochondral bone formation in the setting of various orthopaedic surgical interventions and other skeletal deficiencies.

Keywords

RAR γ ; endochondral bone formation; cartilage; bone defect; BMP

Introduction

Stimulation of bone healing is a challenging, but highly desirable goal for the treatment of various skeletal conditions, including surgical or traumatic bone defects and non-union fractures in the elderly and other compromised patients. Bone morphogenetic proteins (BMPs) and other bone inductive materials -singly or in combination- have been in use for nearly two decades and are clinically proven to improve bone repair^(1,2). However, their use still faces problems including high cost, limited effectiveness and suboptimal repair bone quality^(1,3). In addition, studies have reported that the use of large doses of exogenous BMPs can cause complications such as excess or heterotopic ossification, local inflammation, loosening of fixative devices due to bone resorption, and increased cancer risk^(4,5). Clearly, there is a need to develop additional and safer treatment strategy for bone repair and regeneration.

Retinoic acid (RA), the major active metabolite of retinol, plays important roles in embryonic development, organogenesis, and homeostasis in a variety of tissues and organs⁽⁶⁾. RA is one of key regulators of skeletal patterning during embryogenesis and regulates cartilage and bone development and growth^(7,8). RA acts primarily by binding to its nuclear retinoic acid receptors (RARs). The RARs heterodimerize with retinoid X receptors (RXRs) and regulate cell function through direct activation of target genes and cross-talks with other signaling pathways⁽⁹⁾. Mammals have three RAR isotypes -RAR α , RAR β and RAR γ - and three RXR isotypes -RXR α , RXR β and RXR γ - that form a variety of heterodimer combinations and can exert different functions⁽¹⁰⁾. Overall, the specificity and magnitude of RA action are controlled by the temporospatial patterns of expression of RARs and RXRs and the endogenous cellular levels of RA^(6,11,12).

We previously demonstrated that RAR γ is responsible for retinoid-mediated inhibition of chondrogenesis and that pharmacological stimulation of RAR γ function effectively blocked heterotopic ossification (HO), a pathology in which excess endochondral bone forms and accumulates within muscle and connective tissues⁽¹³⁾. We found that oral administration of selective RAR γ agonists strongly inhibited HO induced by subcutaneous or intramuscular transplantation of rhBMP-2-containing Matrigel or collagen sponge. Furthermore, the agonists reduced HO in mice harboring a constitutive-active Activin receptor-like kinase-2 mutant (ALK2^{Q207D}) similar to ALK2 mutants seen in Fibrodysplasia Ossificans Progressiva (FOP) patients⁽¹³⁾. Given that the RAR γ agonists reduce endochondral bone formation, it follows that RAR γ antagonists should do the opposite and stimulate endochondral bone formation. If so, this would provide a new tool to stimulate bone formation at desired sites. To test these hypotheses, we first determined whether experimental bone healing was affected in *RAR γ* -null mice compared to wild types. We also analyzed whether synthetic RAR γ antagonists stimulated bone formation at ectopic sites and whether they affected and modulated canonical BMP signaling. The data presented here provide strong support for our hypotheses.

Materials and Methods

All animal experiments were performed following institutional guidelines and were approved by the Institutional Animal Care and Use Committees. Animals were housed in a ULAR supervised animal facility with a 12-hour light/dark cycle in a temperature ($22\pm1^{\circ}\text{C}$) and humidity ($55\pm5\%$) controlled room. Animals were provided hygienic animal bedding; all cages contained wood shavings, bedding and cotton pads. The health status of each animal was monitored throughout the experiments by investigators as well as by animal veterinary technicians and veterinarians as per the institutional guidelines. The mice were free of all viral, bacterial, and parasitic pathogens during the experimental schedule. Body weight and activity of the mice indicating their general health were monitored throughout the experiments. RAR γ deficient mice⁽¹⁴⁾ were kindly provided by Dr. Pierre Chambon and Norbert B. Ghyselinck (INSERM, France). Genotyping procedure was described previously⁽⁸⁾. CD1 and C57BL/6J mice were purchased from Charles River (Malvern, PA).

Tibia defect model

Under anesthesia and analgesia, the surgical site was shaved using a sharp razor and a skin incision was made over the medial aspect of the tibia in 2 month-old female RAR γ -null and wild type mice. To gain access to the tibia, the tibialis anterior muscle was divided carefully while avoiding damage to the tibial periosteum. A 1 mm diameter hole was drilled in the midline at about 5 mm below the tibia patellar tendon insertion site, using a 21-gauge needle and dental carbide bur (size 6; Henry Schein, Melville, NY) attached to a micromotor (RAM Products, Dayton, NJ) under continuous irrigation with saline. The skin was closed with 6–0 nylon sutures, and the mice were sacrificed 7 or 14 days after surgery and their tibias were processed for histological and μCT analysis. To assess bone formation activity, some mice received an intraperitoneal injection of 30 mg/kg calcein (C0875; Sigma-Aldrich, St. Louis, MO) 24 h before euthanization⁽¹⁵⁾.

Ectopic bone formation model

Ectopic bone formation was induced in mice as previously described⁽¹⁶⁾. Briefly, recombinant human bone morphogenetic protein-2 (rhBMP-2; Gene Script Corp., Piscataway, NJ) was mixed with growth-factor reduced Matrigel (BD Bioscience, Franklin Lakes, NJ) on ice to a final concentration of 4 $\mu\text{g}/\text{ml}$. Aliquots of 250 μl rhBMP-2-containing Matrigel were injected into left and right subcutaneous abdominal region in 2 month-old CD-1 female mice. Mice were then randomly divided into desired number of groups and received either vehicle control or synthetic retinoid. Ectopic tissue samples were harvested at indicated time points and subjected to μCT and histological analyses.

Preparation and administration of retinoids

R667 (palovarotene, CAS410528-02-8)⁽¹⁷⁾ and NRX 204647⁽¹³⁾ are RAR γ agonists and were synthesized by Atomax Chemicals (Shenzhen, China). RAR γ antagonists CD2665 (CAS 170355-78-9)⁽¹⁸⁾ and MM11253 (CAS345952-44-5)⁽¹⁹⁾ were purchased from Tocris Biosciences (Bristol, UK). RAR γ antagonist 7a was synthesized by Atomax Chemicals (Shenzhen, China) according to the world patent application WO 2005/066115 A2. The concentrations of retinoids used for animal experiments were 1 mg/kg for NRX 204647 and

4 mg/kg for all other compounds unless indicated. Stock solutions of retinoids in DMSO (D2650; Sigma-Aldrich, St. Louis, MO) were stored at -30°C under argon. Before administration, 30 μl aliquots of stock solution were mixed with 70 μl of corn oil (C8267; Sigma-Aldrich, St. Louis, MO) for each dose, and administered to mice by oral gavage using a 20-gauge gavage needle (Fine Science Tools, Foster City, CA) at indicated time points. Vehicle control mice received 30 μl DMSO plus 70 μl corn oil in the same manner.

Histological and immunohistochemical analyses

Tissue samples were fixed in 4% paraformaldehyde overnight at 4°C , decalcified in 10% EDTA (only for calcified samples), processed for paraffin embedding, and serially sectioned at 6 μm -thickness. Sections were stained with hematoxylin and eosin (HE) for general profiling or with alcian blue at pH1.0 and eosin counter staining. For immunohistochemical analysis, antigen retrieval was performed with 0.1% pepsin in 0.02N-HCl for 10 min, and sections were incubated with blocking solution (0.1M NaPB, 1% BSA and 10% normal goat serum) for 1 h and then with primary antibodies against phospho-Smad1/5/8 (#9511; Cell Signaling Technology, Danvers, MA) or anti-RAR γ (#8965; Cell Signaling Technology, Danvers, MA) overnight at 4°C . The antibodies were visualized using rabbit specific HRP/DAB (ABC) detection kit (ab64261; Abcam, Cambridge, MA). After detection, sections were counter stained with 0.5% methyl green. Immunopositive cells are counted in randomly chosen 8 areas ($109 \times 145\mu\text{m}$, 2–3 fields/sample) and the percentage of positive cell number to total cell number was calculated.

Analysis by μCT

After fixation, samples were subjected to μCT analysis using a CT35 scanner (SCANCO USA Inc., Southeastern, PA) at 55 kV and 70 mA for bone volume/tissue volume (BV/TV) analysis. Normalized BV/TV means were calculated by ratios of experimental versus control values.

Immunoblotting

ATDC5 cells were grown until approximately 70% confluence in Dulbecco's modified Eagle's medium/Ham's nutrient mixture F-12 containing 5% fetal bovine serum (FBS). After serum deprivation (0.3% FBS) for 16 h, cultures were treated with rhBMP-2 and/or retinoids for 60 minutes and then lysed in SDS-PAGE sample buffer. Cell lysates were denatured, separated onto 10% SDS-page gel and transferred to PVDF membranes (EMD Millipore, Billerica, MA). Membranes were incubated overnight at 4°C with antibodies against phospho-Smad1/5 (#9616; Cell Signaling Technology, Danvers, MA), phospho-Smad1/5/8 (#9511; Cell Signaling Technology, Danvers, MA), phospho-Erk1/2 (#4370; Cell Signaling Technology, Danvers, MA), or phospho-p38 (#4511; Cell Signaling Technology, Danvers, MA), followed by incubation with horseradish peroxidase (HRP)-conjugated secondary antibody (#7074; Cell Signaling Technology, Danvers, MA). HRP activity was detected using a SuperSignal West Dura Extended Duration Substrate (34075; Thermo Scientific, Waltham, MA). Membranes were re-blotted with antibodies against GAPDH (sc-32233; Santa Cruz Biotechnology, Dallas, TX) for loading normalization.

Reporter assays

Primary mouse epiphyseal chondrocytes were isolated from neonatal C57BL/6J mice as described previously⁽²⁰⁾ and were plated on 48 or 96 well plates and serum-starved overnight. Cultures were co-transfected with canonical BMP signaling reporter Id1-Luc⁽²¹⁾ or retinoic acid response element (RARE)-Luc⁽¹²⁾ plasmids and phRG-TK (E2241; Promega, Madison, WI) using Lipofectamine LTX and Plus reagent (#15338100; Life Technologies, Grand Island, NY) according to the manufacturer's protocol. Twenty four hours later, cells were treated with indicated concentrations of rhBMP-2 and/or retinoid and incubated for an additional 24 hours. Cells were then harvested and subjected to dual luciferase assay (E1960; Promega, Madison, WI). Firefly luciferase activity was normalized to renilla luciferase activity generated by phRG-TK.

Statistical analysis

All results were examined by one-way factorial ANOVA followed by Dunnett's or Bonferroni's post hoc multiple comparison tests (Prism 5; GraphPad Software, La Jolla, CA). *p*-values less than 0.01 were considered as statistically significant versus control (**p*<0.01, as indicated by brackets).

Results

A larger volume of cartilage tissues are induced in bone defect in *RARγ*-null mice

We showed previously that treatment with *RARγ*-selective agonists inhibits ectopic endochondral bone formation in mouse models of HO, suggesting that the function of *RARγ* is amenable to experimental manipulation and could be manipulated to different ends⁽¹³⁾. To test this interesting possibility, we asked whether *RARγ* deficiency would actually stimulate formation of endochondral bone. We drilled a 1 mm round-shaped hole in the cortical bone in the upper 1/3 region of tibia (Fig. 1A) in adult wild-type and *RARγ*-null mice and monitored the repair process over time. In control mice at day 7 from surgery, the bone defect was filled with fibrous tissues containing a small amount of cartilaginous nodules that were positive for alcian blue staining (Fig. 1B). In contrast, we found abundant cartilage tissue around the perimeter of the defect in companion *RARγ*-null mice (Fig. 1E). Analyses by μ CT and calcein labeling at 2 weeks after surgery revealed that the *RARγ*-null mice (Fig. 1F and G) exhibited higher bone forming activity compared to wild type mice (Fig. 1C and D). To assess more closely the degree of cartilage formation, we prepared serial sagittal sections through the tibias, stained them with alcian blue and eosin (Fig. 1H) and quantified the cartilaginous regions that had developed within and around the defect (Fig. 1I). The overall areas of cartilaginous tissue were significantly larger in *RARγ*-null mice than control mice at 1 week from surgery (Fig. 1H), but dwindled over time as endochondral ossification progressed and cartilage was replaced by bone (Fig. 1I). We did not quantify bone itself because it was difficult to distinguish repair from endogenous bone in tissue sections. The above data indicate that absence of *RARγ* leads to more exuberant cartilage formation in response to bone defect.

RAR γ antagonists stimulate ectopic endochondral bone formation

To further evaluate the roles of RAR γ and the efficacy of RAR γ antagonists on bone formation, we employed an efficient mouse ectopic bone model⁽¹⁶⁾ and compared the effects of three selective RAR γ antagonists -CD2665, MM11253, and 7a- with those of the RAR γ agonist NRX204647 (Fig. 2A). We subcutaneously injected 250 μ l aliquots of Matrigel containing 1 μ g of rhBMP-2 at two ventral subdermal sites in 2 month-old CD-1 mice. In the line with our previous studies, the injected BMP-Matrigel mixture rapidly solidified and elicited formation of endochondral bone within 10 to 12 days (Fig. 2B)⁽¹⁶⁾. The BMP-Matrigel injected mice were randomly subdivided into groups and subjected to oral gavage with vehicle corn oil, RAR γ agonist or each of the 3 antagonists on day 3, 5 and 7. Ectopic tissue masses were collected on day 12 and subjected to X-ray and μ CT analyses. As expected, the RAR γ agonist NRX204647 strongly inhibited bone formation compared to vehicle-treated control mice as determined by μ CT analysis (Fig. 2C). In contrast, each of the three RAR γ antagonists had significantly increased the amount of detectable mineralized bone (Fig. 2C and D). Because the three antagonists represent different backbone structures characteristic of this drug class (Fig. 2A), the data suggest a stimulatory drug class effect by these retinoids. There was no significant difference in bone density between vehicle and antagonist-treated groups (data not shown). To verify specificity of drug action, we carried out similar ectopic bone formation experiments with RAR γ -null mice and treated them with vehicle or one of the antagonists (CD2665) (Fig. 2E). We observed no significant differences in bone formation in vehicle- versus drug-treated mice, indicating that drug action was isotype-specific and that antagonizing endogenous RAR γ function is indeed required to increase ectopic bone formation (Fig. 2E).

To clarify the mode of action of RAR γ antagonists on successive phases of the endochondral bone formation process, we carried out closer histological and histomorphometric analyses on ectopic tissue samples collected at different time points from control and RAR γ antagonist-treated mice. By day 3 in control mice, the BMP-Matrigel scaffold already contained an appreciable number of invading host cells that likely included skeletal progenitors and inflammatory cells (Fig. 3A, B). By day 6, a significant amount of alcian blue-positive cartilage was presented (Fig. 3A) that had increased significantly by day 9 and was being replaced by bone and marrow over time and almost completely by day 20 (Fig. 3A). Histomorphometric analyzes of ectopic tissue masses from companion mice treated with RAR γ antagonist CD2665 as above and carried out along the three topographical planes indicated in Fig. 3B showed that the tissue masses contained a significantly higher amount of cartilage than controls on day 7 (Fig. 3C Cartilage Day 7). While the overall area of cartilage had decreased by day 12, it was still more abundant in treated- than control samples (Fig. 3C). Analysis by μ CT showed that the mineralized tissue volume was about 2-fold higher in RAR γ antagonist-treated than control samples on day 12 (Fig. 3C). These findings indicated that the RAR γ antagonists stimulate cartilage formation, representing the early phase of endochondral bone formation and resulting in increased bone formation at later stages. To examine these later stages more closely, mice implanted with BMP-Matrigel were treated with a RAR γ antagonist on day 7, 9 and 11, and the ectopic tissue masses were analyzed on day 12 or 17 (Fig. 3D). On day 12, we already observed more cartilaginous tissue in RAR γ antagonist-treated than control samples (Fig. 3D). The

amount of mineralized tissue appeared to be increased as well, but the difference was not significant over control values (Fig. 3D). By day 17, the cartilaginous tissue area was reduced in both groups, but $RAR\gamma$ -antagonist treated samples displayed a significantly larger volume of bone than controls (Fig. 3D). The findings reaffirm that $RAR\gamma$ antagonists enhance cartilage formation during the middle stages of the endochondral ossification process and eventually lead to increases in bone formation.

Inhibition of $RAR\gamma$ function enhances phosphorylated Smad1/5/8 levels

Studies have indicated that there are reciprocal antagonistic relationships between the retinoid and BMP signaling pathways in several biological systems^(11,22–24), and our own studies suggested that the suppression of chondrogenesis and heterotopic ossification by $RAR\gamma$ agonists involved inhibition of canonical BMP signaling⁽¹³⁾. Thus, it became important to determine if and how BMP signaling was affected in the various experimental approaches above. First, we determined the distribution of phosphorylated Smad1/5/8-positive cells within the regenerating tibial bone defect areas in wild type and $RAR\gamma$ -null mice (Fig. 4A). There were far more numerous and widely distributed pSmad1/5/8-positive cells within the bone repair region in mutant than control mice on day 7 from surgery (Fig. 4A). In an analogous manner, pSmad1/5/8-positive cells were more numerous and obvious in ectopic BMP-Matrigel tissue masses from mice treated with $RAR\gamma$ antagonist CD2665 than controls on day 5 after implantation (Fig. 4B). This increase was evident for cells seemingly migrating into the scaffold as well as those surrounding it (Fig. 4B). Semi-quantitative analysis of pSmad1/5/8 positive cells confirmed the observation (Fig. S1). Thus, canonical BMP signaling is increased by deficiency of $RAR\gamma$ or by treatment with $RAR\gamma$ antagonists. We also confirmed broadened expression of pSmad1/5/8 in $RAR\gamma$ -null growth plate compared to that of wild type (Fig. 4C), suggesting that similar antagonistic relationship exist also in the endochondral ossification process during development.

We next examined distribution of endogenous $RAR\gamma$ in cartilage and in cells within the ectopic BMP-Matrigel tissue masses (Fig. 5). $RAR\gamma$ is readily detectable and likely abundant in growth plate chondrocytes (Fig. 5A and C), but was undetected in $RAR\gamma$ -null mice as expected (Fig. 5B and D). Detection of $RAR\gamma$ was mostly observed in chondrocytes within the BMP-Matrigel tissue masses (Fig. 5E and F).

$RAR\gamma$ antagonists enhance Smad phosphorylation and *Id1* reporter activity

Lastly, we examined whether $RAR\gamma$ has cell autonomous roles in BMP signaling in chondrocytes. Chondrocytes freshly isolated from neonatal mouse cartilaginous long bone epiphyses were treated in culture with increasing concentrations of $RAR\gamma$ antagonists (CD2665 or MM11253) for 24 hours in the presence of 10 ng/ml rhBMP2 and subjected to immunoblot analysis. The pSmad1/5 levels were significantly enhanced by $RAR\gamma$ antagonist treatment at each dose tested (Fig. 6A, pSmad1/5). BMP2 is also known to stimulate the MAPK pathways⁽²⁵⁾. Treatment with $RAR\gamma$ antagonists moderately increased the levels of phosphorylation p38, and did not affect Erk1/2 phosphorylation appreciably (Fig. 6A, pp38 and pErk1/2). Enhancement of BMP-Smad signaling by $RAR\gamma$ antagonists was confirmed by *Id1-luc* reporter assays. The activity of this reporter was slightly increased by $RAR\gamma$ antagonist treatment in absence of exogenous rhBMP-2, but was significantly

enhanced in antagonist treatment (Fig. 6B). Specificity of responses was confirmed by the fact that the RAR γ antagonists inhibited *RARE-luc* reporter activity (Fig. 6C).

Discussion

The present study provides multiple lines of evidence that RAR γ is an important and novel regulator of endochondral bone formation. We observed appearance of cartilaginous tissue in the bone defect followed by replacement with mineralized materials and that the volume of the cartilaginous tissues are much higher in RAR γ -null mice compared to control mice. The healing process after the bone defect injury likely involves both endochondral ossification and intramembranous ossification although definitive histological delineation of the two events is difficult. Induction of a larger volume of cartilaginous tissues in RAR γ -null mice suggests that a lack of RAR γ signaling stimulates endochondral ossification. However, the absence of RAR γ signaling might lead a shift of the healing process from the intramembranous mode to the endochondral mode. Furthermore, we demonstrated that pharmacological inhibition of RAR γ function by selective antagonist treatment stimulates ectopic endochondral bone formation. In both experimental models, local BMP signaling is enhanced as determined by marked increases in pSmad1/5/8 levels. This finding is in line with our in vitro observations suggesting that treatment with RAR γ antagonists stimulates BMP signaling rapidly and significantly. Together with our previous studies showing that pharmacological activation of RAR γ function blocks ectopic endochondral ossification in mice and inhibits canonical BMP signaling, the results presented here indicate that RAR γ is a critical regulator and modulator of endochondral bone formation and represents a plausible and powerful therapeutic target to manipulate and enhance bone formation in various pathological skeletal conditions.

Which are the cell types that are targeted by RAR γ antagonists and elicit the stimulation in ectopic endochondral bone formation? Our data on endogenous gene expression patterns of RAR γ protein suggest that a primary target are chondro-progenitor cells and chondrocytes, but not osteoblasts (Figure 5). This is supported by previous findings that retinoid pan-antagonists stimulate chondrogenesis of undifferentiated mesenchymal cells in high-density culture⁽²⁶⁾ and that unliganded RAR γ supports, and is required for, cartilage matrix production in chondrocytes⁽⁸⁾. Thus, inhibition of RAR γ action by the retinoid antagonists is most likely able to stimulate chondrogenic differentiation of mesenchymal cells and to increase matrix production in chondrocytes, resulting in increases in ectopic cartilage volume and in turn, stimulation of bone formation. Minegishi et al. reported that Cyp26b1, an enzyme that degrades (oxidizes) RA, is expressed in cartilage and that ablation of Cyp26b1 caused marked reduction of cell proliferation in growth plate⁽²⁷⁾, indicating that reduction of RA levels is important for proliferation of chondrocytes. This mechanism may also have facilitated formation and growth of ectopic cartilage under RAR γ antagonist treatment regimens used here.

Increases and decreases in endochondral bone formation respectively elicited by RAR γ antagonists or agonists are accompanied by reciprocal changes in BMP signaling. This is likely to be the key mechanism of RAR γ action as BMPs and BMP signaling normally play indispensable roles in endochondral ossification⁽¹³⁾. RAR γ modulates cellular function

through regulation of target genes (genomic action) as well as via interactions with other signaling pathways (non-genomic action)^(9,28). In our studies, changes in phosphorylation levels of Smad proteins were observed within 1 hour after RAR γ agonist or antagonist treatment in cultured cells. This rapid change indicates that the effects are at least partly mediated by the non-genomic function of RAR γ rather than through genomic action that would require changes in gene expression and production of new proteins, both likely requiring more than 1 hour. *In vitro* studies showed that RAR γ physically interacts with the MH2 domain of Smad3 and down-regulates Smad3/4-dependent transcription in the presence of RAR γ agonists, while Smad3/4-dependent transcription activity was markedly increased by RAR pan-antagonist treatment⁽²⁹⁾. The MH2 domain is used for protein-protein interactions, including hetero-trimerization of Smads, and is highly conserved especially among the R-Smad proteins⁽³⁰⁾. Thus, RAR γ may directly interact with Smad1/5/8 proteins and modulate their function. On the other hand, RAR γ is also known to interact with Src, a non-receptor tyrosine kinase, and ligand-bound RAR γ enhances Src kinase activity⁽³¹⁾. Other reports have demonstrated that Src interacts with the C-terminal domain of BMP type II receptors⁽³²⁾, and suggest that RAR γ indirectly interacts with BMP receptor and could modulate BMP receptor function. It is also possible that RAR γ regulates the transcription of genes related to BMP signaling and such mechanism should be studied. Further characterization of RAR γ -BMP signaling interactions would not only be important for understanding molecular mechanisms and circuits, but would also be relevant to drug development for diseases involving dysregulation of BMP signaling, such as pulmonary arterial hypertension (PAH)⁽³³⁾, Fibrodysplasia Ossificans Progressiva⁽³⁴⁾ and certain types of cancer⁽³⁵⁾.

Synthetic retinoids are small in size (~400MW), relatively stable and easy to administer^(36,37). RAR γ antagonists and agonists may thus be easy-to-use and versatile modulators of endochondral bone formation in various clinical settings. For example, large bone defects caused by trauma or surgery are currently treated by local application of growth factors, such as BMP2 and BMP7, scaffolds or bone grafts, adult stem/progenitor cells or their combinations^(1,2,38). RAR γ antagonists could be applied to the defects by incorporation into the scaffolds, by oil-based ointments or oral administration. Since the RAR γ antagonists enhance BMP signaling only in RAR γ responsive cells, it may effectively promote endochondral ossification and reduce the amount of recombinant BMPs needed to repair a bone defect. Such a reduction would not only reduce cost, but may decrease the risk of adverse effects by the BMPs such as local inflammation, tissue swelling, osteolysis, and so on. Spinal fusion surgery is another possible situation in which RAR γ antagonists or agonists may be useful. Since the Food and Drug Administration first approved the use of rhBMP-2 for fusion of lumbar spine in adult patients in 2002, BMPs have been widely used in connection with spinal fusion surgeries. Although the BMPs are beneficial in this setting, they can cause complications such as heterotopic ossification, neuropathy and compression of airway in some patients⁽³⁹⁾. These adverse effects are difficult to predict due to variability in patient-to-patient BMP responses. Thus, the RAR γ antagonists may reduce the risk of these complications by reducing the BMP amounts needed for spinal fusion surgery. Ongoing experiments are testing these important and clinically-relevant possibilities.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

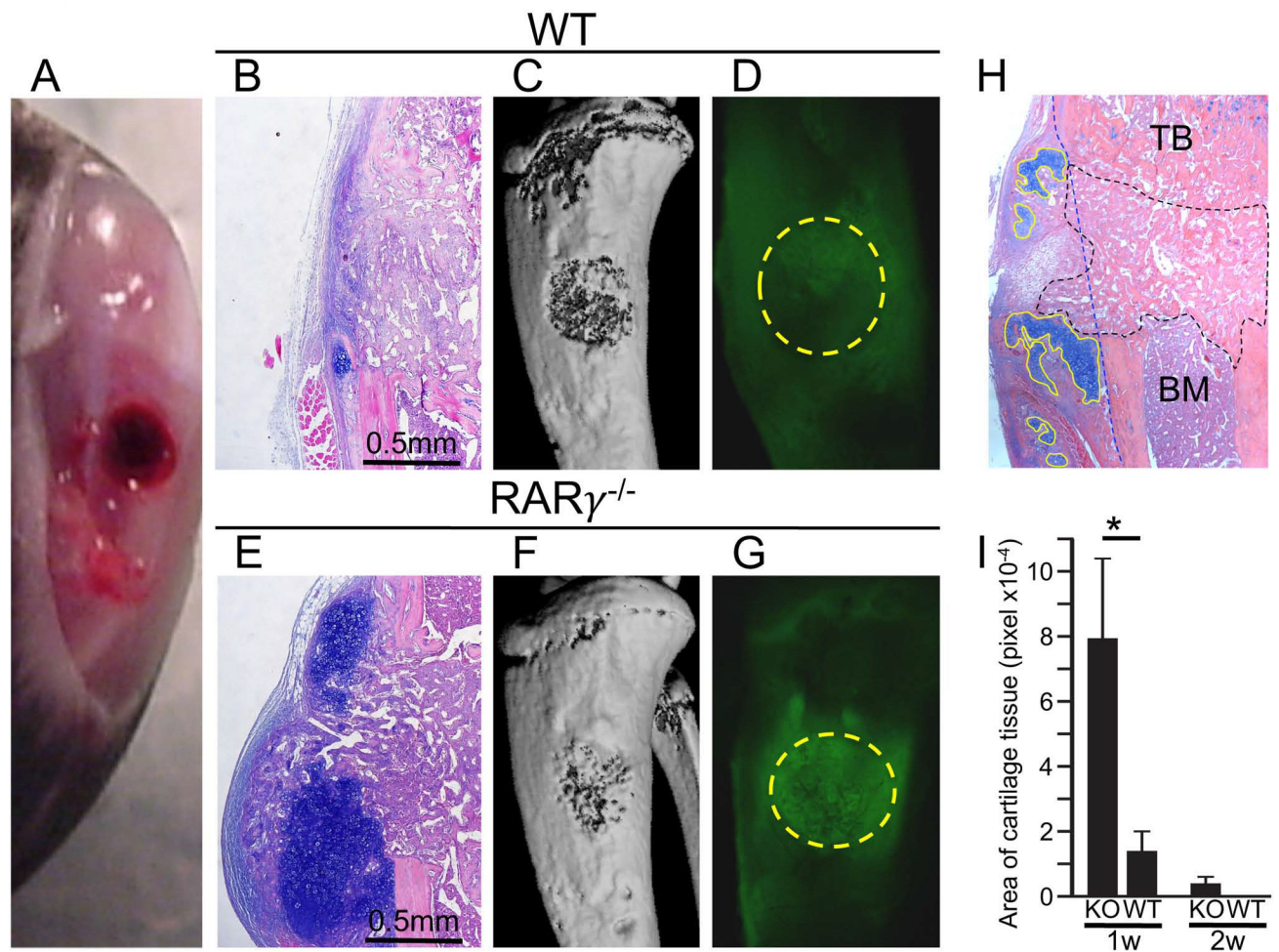
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**Fig. 1.**

A larger volume of cartilaginous tissue was formed in $RAR\gamma$ -null than wild type mice in tibial bone defect model. (A) A round shaped hole was made on the medial aspect of adult mouse tibias using a 22 gauge needle and dental carbide bur. (B and E) Alcian blue and eosin staining of tibia sections on day 7 from surgery. (C and F) μ CT images of tibias on day 14 post-surgery. (D and G) Fluorescent macroscopic images of calcein-labeled tibias on day 14 post-surgery. Yellow dotted lines indicate the location of the bone defects. (H) Representative image of sections used for quantitative analyses of cartilage tissue associated with the bone defect. Alcian blue positive cartilaginous tissue is highlighted by the yellow line and was quantified using pixel numbers and Image J software. TB: trabecular bone; BM: bone marrow. (I) Histograms depicting the areas of alcian blue-positive cartilage within the defects in wild type (WT) and $RAR\gamma$ -null (KO) mice at 1 and 2 weeks from surgery. $N=4$, $*P<0.01$

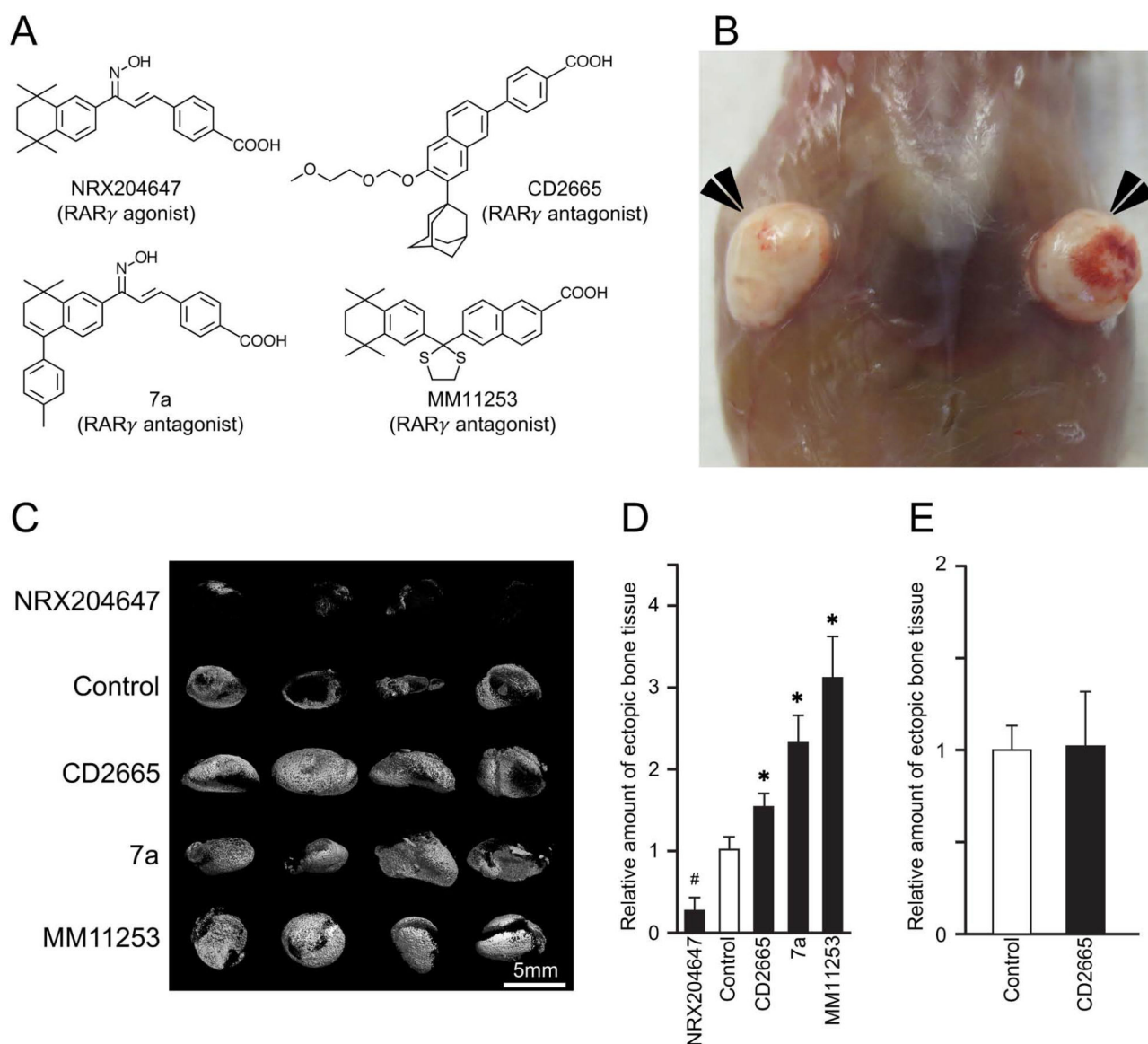


Fig. 2. Selective RAR γ antagonists enhance ectopic bone formation. (A) Chemical structures of the synthetic RAR γ agonist and antagonists used in this study. (B) Whole mount images showing the ectopic BMP-Matrigel masses on day 12. Arrowheads point to blood vessels reflecting ongoing endochondral ossification. (C) μ CT images of ectopic tissue masses harvested on day 12 from Matrigel injection. Four repeat samples from each treatment group are shown. (D) Histograms showing the calculated amounts of ectopic bone volume expressed as BV/TV. Note that while the RAR γ agonist NRX204647 significantly reduced BV/TV ($\#p < 0.01$), each of the antagonists tested significantly increased it over control values (white histogram) ($*p < 0.01$). The total number of samples used for these analyses were: 8 for NRX204647; 31 controls; 26 for CD2665; 8 for 7a; and 6 for MM11253. The results were combined from 4 independent experiments. (E) Histograms showing that

CD2665 treatment had no effect on ectopic bone formation in *RAR γ* -null mice. Y-axis indicates relative amount of bone compared to control. *N*=8

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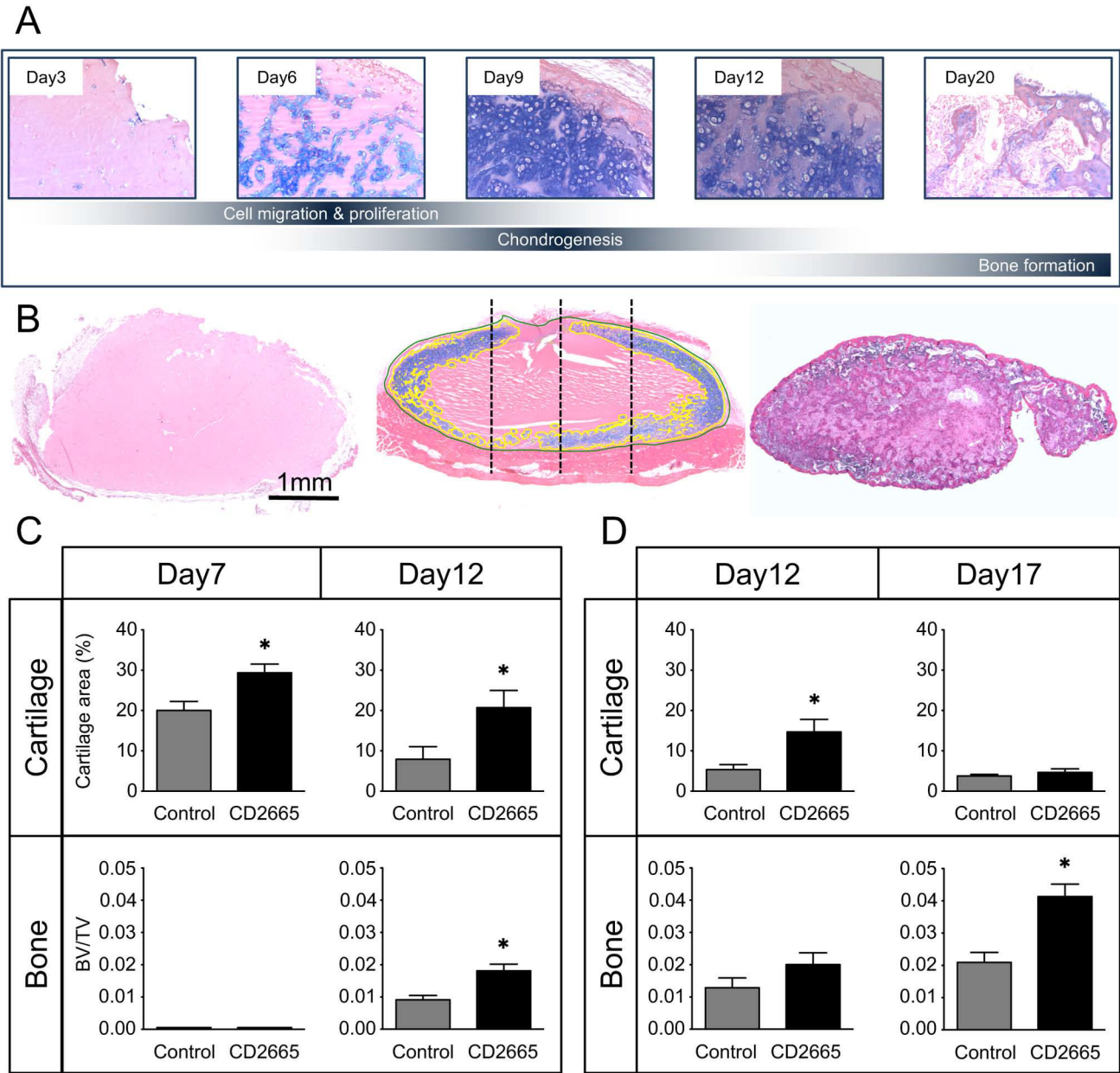


Fig. 3. RAR γ antagonists differentially modulate cartilage and bone formation. (A) Histological images of Alcian blue-stained sections of ectopic masses collected at 3, 6, 9, 12 and 20 days from BMP-Matrigel implantation and showing initial cell immigration and progressive cartilage and bone formation over time. (B) Histology of whole ectopic masses stained with alcian blue (left and middle) or H-E (right). Images from left to right are ectopic masses of day 3, day 9, and day 20. (C and D) Quantification of cartilage and bone in ectopic masses from control and RAR γ antagonist-treated mice on indicated times. Samples were first subjected to pCT analysis and were then sectioned and stained with alcian blue for quantification of cartilage. For each group, 3 sections each with 1 mm interval (example is shown in dotted vertical lines in panel B) of 4 independent samples were analyzed. For the

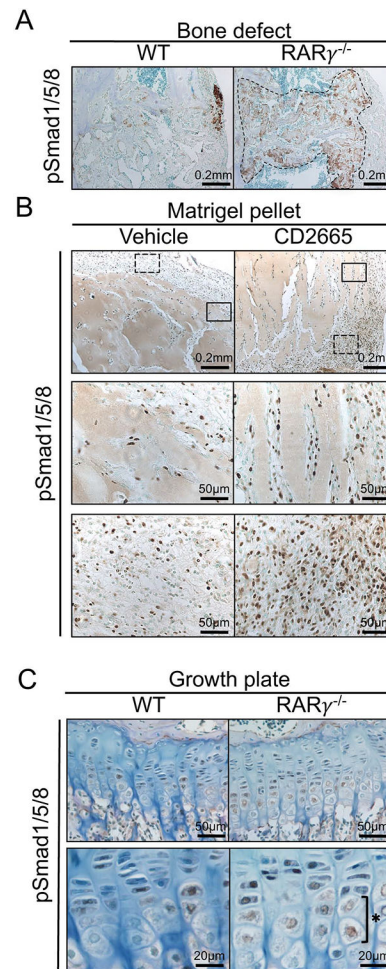
experiments shown in (C), mice were gavaged with vehicle or 4 mg/kg of CD2665 on day 3, 5 and 7 from BMP-Matrigel implantation and samples were collected on day 7 or 12. For experiments in (D), mice were gavaged with vehicle or 4 mg/kg CD2665 on day 7, 9 and 11 from implantation and analyzed on day 12 or 17.

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**Fig. 4.**

Inhibition of $RAR\gamma$ function increases the number of pSmad1/5/8-positive cells. (A) Immunohistochemical detection of pSmad1/5/8-positive cells within bone defects of wild type (WT) and $RAR\gamma$ -null ($RAR\gamma^{-/-}$) tibias 1 week after surgery. (B) Distribution of pSmad1/5/8-positive cells within ectopic BMP-Matrigel masses on day 5 from implantation. Following implantation, mice received vehicle corn oil (Vehicle) or 4 mg/kg CD2665 on day 1 and 3, and ectopic samples were collected on day 5 and subjected to pSmad staining. Top panels are lower magnified images. Middle and bottom panels are higher magnification of regions indicated by the solid (middle) and dotted (bottom) lines. (C) Expression patterns of pSmad1/5/8 proteins in tibial growth plates of wild type (WT) and $RAR\gamma$ -null ($RAR\gamma^{-/-}$) mice. Asterisk indicates expanded expression of pSmad proteins in the hypertrophic zone of $RAR\gamma$ -null growth plate.

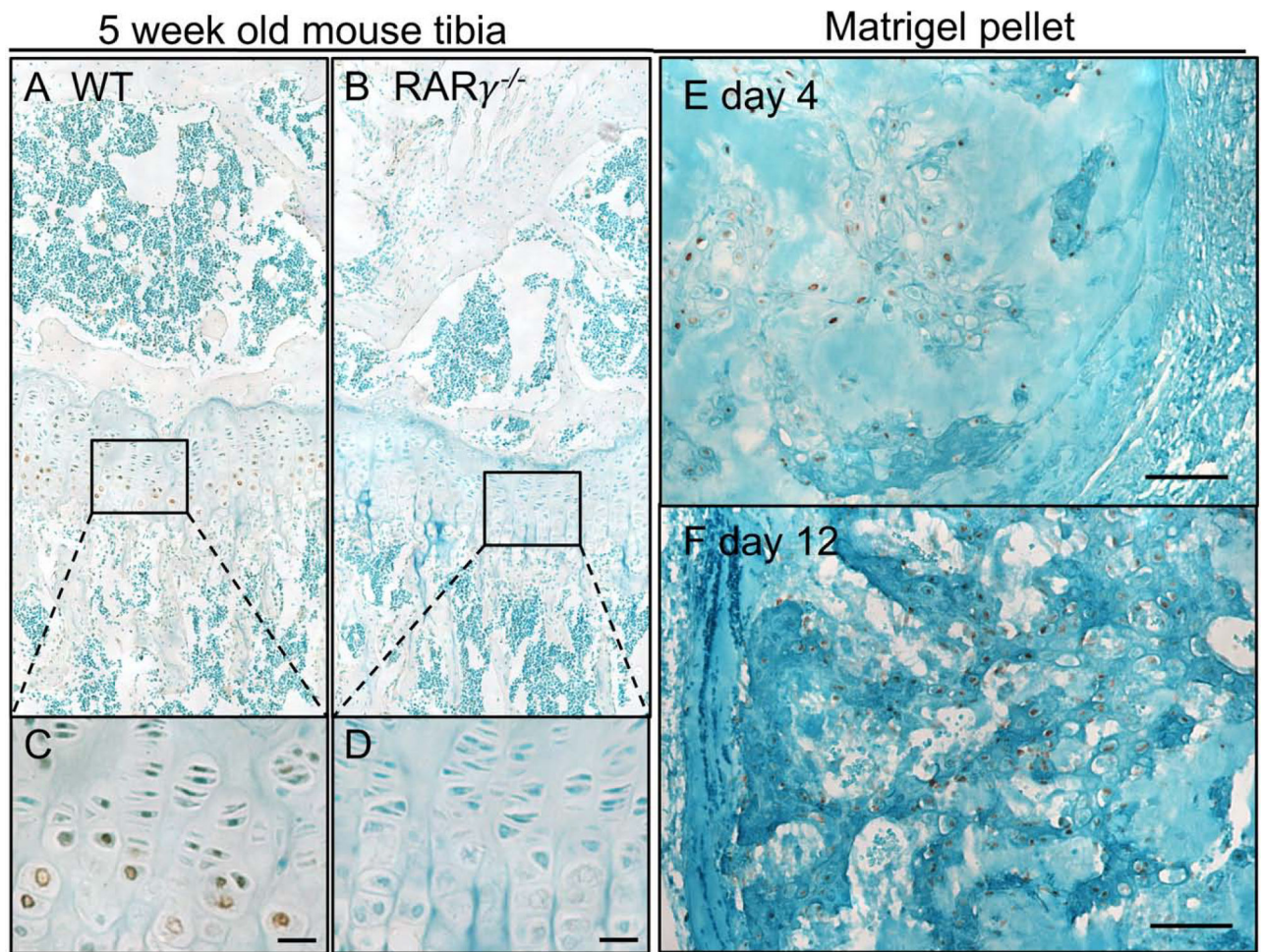
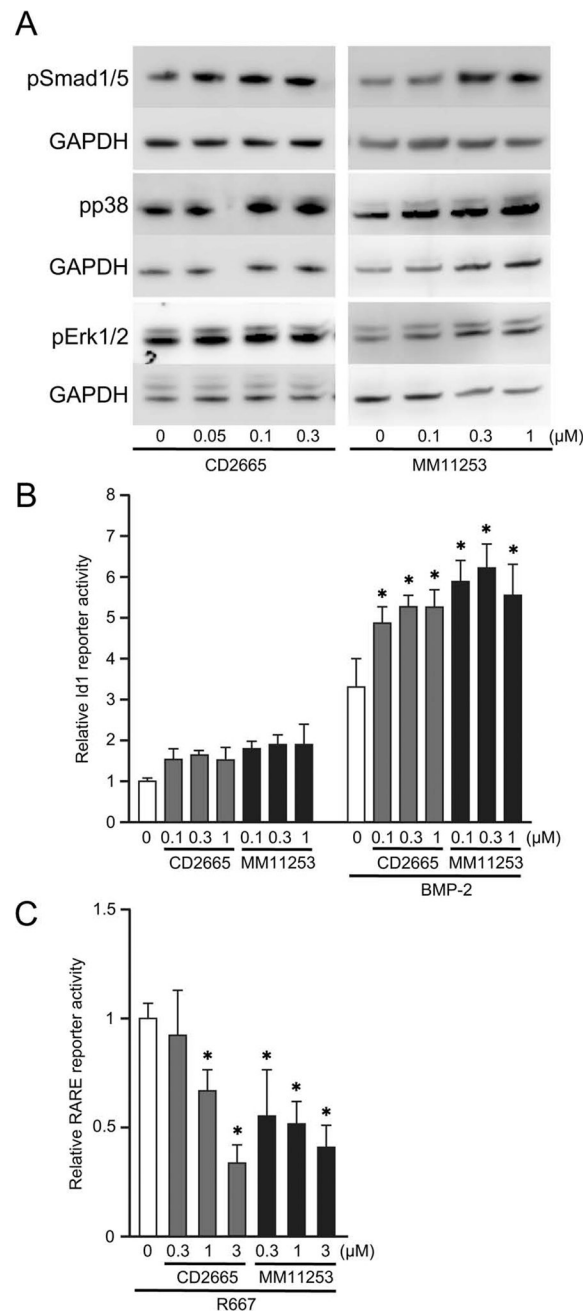


Fig. 5.

Expression of $RAR\gamma$ in growth plate chondrocytes and cells in BMP-Matrigel masses. (A, B) Immunohistochemical detection of $RAR\gamma$ in longitudinal sections of tibial growth plate in 5-weeks old wild-type (A) and $RAR\gamma$ -null mice (B). (C) and (D) are higher magnification of the indicated area in (A) and (B). Bar = 100 μ m. (E and F) Immunohistochemical detection of $RAR\gamma$ in BMP-Matrigels 4 (E) and 12 (F) days after implantation. Bar=100 μ m

**Fig. 6.**

RAR γ antagonists enhance BMP signaling in ATDC5 cells and primary chondrocytes. (A) Effects of RAR γ antagonist treatment on levels of phosphorylated Smad1/5, p38 and Erk1/2 in the presence of retinoid antagonists and rhBMP-2. Primary chondrocytes were seeded at a density of 4×10^4 cells/16mm well and maintained in 0.3% FBS DMEM overnight. Cells were then treated with indicated amounts of CD2665 or MM11253 for 1h in the presence of 10ng/ml rhBMP-2. Levels of phosphorylated Smad1/5, p38 and Erk1/2 were analyzed by immunoblotting. Membranes were re-blotted with anti-GAPDH antibody for normalization. (B) Effects of RAR γ antagonists on the activity of *Id1-luc*, a canonical BMP receptor

signaling reporter. ATDC5 cells were treated with indicated doses of CD2665 or MM11253 in the absence or presence of 10ng/ml rhBMP-2. (C) Monitoring activity of RAR γ antagonists by RARE reporter assay. Cells were treated with indicated doses of CD2665 or MM11253. To detect the effect of RAR γ antagonists in retinoid deficient culture condition (0.3% FBS DMEM), 10nM RA was added to all cultures in experiments A, B and D.

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